



## Yakeen Plus (2025)

### Practice Test - 16

**DURATION : 200 Minutes**

**DATE : 19/12/2024**

**M. MARKS : 720**

#### Topics Covered

|                    |                                                                                                                                                                                                           |
|--------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Physics :</b>   | Electrostatics, Current Electricity, Magnetic Effects of Current and Magnetism, Electromagnetic Induction and Alternating Currents, Electromagnetic Waves                                                 |
| <b>Chemistry :</b> | Thermodynamics & Thermochemistry, Chemical Kinetics, Alcohols, Ethers and Phenols, Aldehydes, Ketones and Carboxylic Acids, The p-Block Elements (Group 13,14,15, 16, 17, 18), The d and f-Block Elements |
| <b>Biology :</b>   | <b>(Botany):</b> Sexual Reproduction in Flowering Plant, Principle of Inheritance and Variation, Microbes in Human Welfare<br><b>(Zoology):</b> Human Reproduction, Reproductive Health, Evolution        |

#### General Instructions:

1. Immediately fill in the particulars on this page of the test booklet.
2. The test is of **3 hours 20 min.** duration.
3. The test booklet consists of **200** questions. The maximum marks are **720**.
4. There are **four Section** in the Question Paper, **Section I, II, III & IV** consisting of Section-I (Physics from question number **1 to 50**), Section-II (Chemistry: from question number **51 to 100**), Section-III (Botany: from question number **101 to 150**) & Section IV (Zoology: from question number **151 to 200**), having 50 Questions in each Subject and each subject is divided in two Section, Section A consisting 35 questions (all questions are compulsory) and Section B consisting 15 Questions (Any 10 questions are compulsory).
5. There is only one correct response for each question.
6. Each correct answer will give 4 marks while 1 Mark will be deducted for a wrong MCQ response.
7. No student is allowed to carry any textual material, printed or written, bits of papers, pager, mobile phone, any electronic device, etc. inside the examination room/hall.
8. On completion of the test, the candidate must hand over the Answer Sheet to the Invigilator on duty in the Room/Hall. However, the candidates are allowed to take away this Test Booklet with them.

## Answer Key

Q1 (4)  
Q2 (2)  
Q3 (2)  
Q4 (4)  
Q5 (2)  
Q6 (4)  
Q7 (1)  
Q8 (1)  
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Q67 (3)  
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Q200 (2)



## Hints &amp; Solutions

Note: scan the QR code to watch video solution

**Q1 Text Solution:**

(D)

$$\vec{B} = \vec{B}_1 + \vec{B}_2 + \vec{B}_3$$

$$\vec{B} = 0 + \frac{\mu_0 i}{4\pi r} \frac{\pi}{2} \otimes + \frac{\mu_0 i}{4\pi r} \odot$$

$$B = \frac{\mu_0 i}{4\pi r} \left( \frac{\pi}{2} - 1 \right)$$

**Q2 Text Solution:**

(B)

Resistance of first bulb,

$$R_1 = \frac{V_1^2}{P_1} = \frac{250 \times 250}{100} = 625 \Omega$$

Resistance of second bulb,

$$R_2 = \frac{V_2^2}{P_2} = \frac{200 \times 200}{100} = 400 \Omega$$

In a series circuit, potential divides in proportion to resistance. Potential drop across bulb  $B_2$  is

$$\therefore V'_2 = \frac{R_2}{R_1 + R_2} \times V = \frac{400}{625 + 400} \times 250 \text{ V}$$

$$= 97.56 \text{ V} \approx 98 \text{ V}$$

**Q3 Text Solution:**

(B)

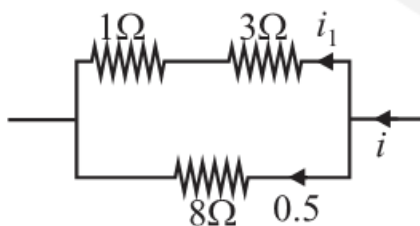
For balanced Wheatstone bridge,

$$\frac{R_1}{R_2} = \frac{R_3}{R_4}, \frac{2}{10} = \frac{R_{\text{eff}}}{4}, R_{\text{eff}} = 0.8 \Omega, \frac{1}{0.8}$$

$$= \frac{1}{4} + \frac{1}{R}$$

On solving  $R = 1 \Omega$ **Q4 Text Solution:**

(D)



$$2 = i' \cdot 8$$

$$i' = \frac{1}{2} \text{ A} = 0.5 \text{ A}$$

$$\frac{0.5}{i_1} = \frac{4}{8}$$

$$i_1 = 1 \text{ A}$$

$$P = 3 \text{ W}$$

**Q5 Text Solution:**

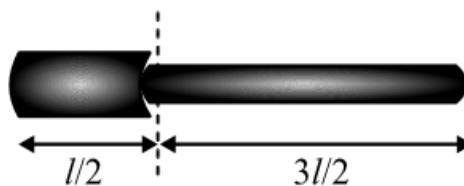
(B)

$$B = B_1 + B_2 = \frac{\mu_0 i}{4r_1} + \frac{\mu_0 i}{4r_2}$$

$$B_{\text{total}} = \frac{\mu_0 I}{4} \left( \frac{r_1 + r_2}{r_1 r_2} \right)$$

**Q6 Text Solution:**

(D)



Resistance of left part = Resistance of right part

$$R_1 = \frac{\rho l/2}{A} = \frac{\rho l}{2A}$$

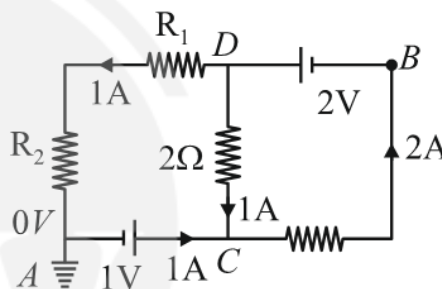
$$R_2 = \frac{\rho 3l/2}{A/3} = \frac{9\rho l}{2A}$$

Net resistance

$$R' = \frac{\rho l}{2A} + \frac{9\rho l}{2A} = \frac{5\rho l}{A} = 5R$$

**Q7 Text Solution:**

(A)



By applying KVL along path ACDB

$$V_A + 1 + 2 \times 1 - 2 = V_B \Rightarrow V_B = 1\text{V}$$

**Q8 Text Solution:**

(A)

$$P_{\text{Total}} = \frac{V^2}{R_{\text{eq}}} \text{ So } R_{\text{eq}} = \frac{V^2}{P_{\text{Total}}} = \frac{10 \times 10}{30} = \frac{10}{3}$$

$$\text{As } R_{\text{eq}} = \frac{R_1 R_2}{R_1 + R_2} \Rightarrow \frac{10}{3} = \frac{5R}{5+R} \Rightarrow R = 10 \Omega$$

**Q9 Text Solution:**

(A)

$$\text{Induced current in the circuit } i = \frac{Bvl}{R}$$

Magnetic force acting on the wire

$$F_m = Bil = B \left( \frac{Bvl}{R} \right) l$$

$\Rightarrow F_m = \frac{B^2 v l^2}{R}$  External force needed to move the rod with constant velocity

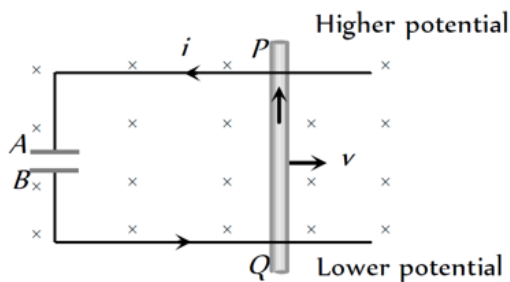
$$(F_m) = \frac{B^2 v l^2}{R} = \frac{(0.15)^2 \times (2) \times (0.5)^2}{3}$$

$$= 3.75 \times 10^{-3} \text{ N}$$

**Q10 Text Solution:**

(A)





$Q = CV = C(Bvl) = 10 \times 10^{-6} \times 4 \times 2 \times 1 = 80 \mu\text{C}$   
According to Fleming's right hand rule induced current flows from Q to P. Hence P is at higher potential and Q is at lower potential. Therefore A is positively charged and B is negatively charged.

**Q11 Text Solution:**

(B)

Due to magnetic field, wire will experience an upward force

$$F = Bil = B \left( \frac{Bvl}{R} \right) l \Rightarrow F = \frac{B^2 vl^2}{R}$$

If wire slides down with constant velocity, then

$$F = mg \Rightarrow \frac{B^2 vl^2}{R} = mg \Rightarrow v = \frac{mgR}{B^2 l^2}$$

**Q12 Text Solution:**

(C)

At the time  $t = 0$ ,  $e$  is max and is equal to  $E$ , but current  $i$  is zero.

As the time passes, current through the circuit increases but induced emf decreases.

**Q13 Text Solution:**

(B)

$$P = Fv = Bil \times v = B \left( \frac{Bvl}{r} \right) l \times v = \frac{B^2 v^2 l^2}{r}$$

$$\Rightarrow P \propto v^2$$

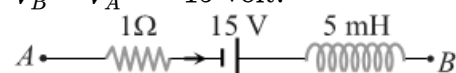
**Q14 Text Solution:**

(C)

By using Kirchoff's voltage law

$$V_A - iR + E - L \frac{di}{dt} = V_B$$

$$V_B - V_A = 15 \text{ volt.}$$



**Q15 Text Solution:**

(B)



$$\varepsilon_{\text{ind}} = \frac{B_0 \omega l^2}{2}$$

$$\varepsilon_{\text{ind}} = \frac{B_0 \omega (2R)^2}{2}$$

$$\varepsilon_{\text{ind}} = 2B_0 \omega R^2$$

**Q16 Text Solution:**

(B)

Two inductors are in parallel. Therefore,

$$V_1 = V_2$$

$$\text{or } L_1 \left( \frac{di_1}{dt} \right) = L_2 \left( \frac{di_2}{dt} \right)$$

$$\text{or } L_1 (di_1) = L_2 (di_2)$$

Integrating, we get

$$L_1 i_1 = L_2 i_2$$

$$\frac{i_1}{i_2} = \frac{L_2}{L_1}$$

**Q17 Text Solution:**

(D)

For inductor the voltage leads the current, for

$$t = 0 \text{ to } t = \frac{T}{2}$$

$$V = L \frac{di}{dt} = L \frac{d}{dt} \left( \frac{2I_0 t}{T} \right) = \text{constant}$$

$$\text{for } t = \frac{T}{2} \text{ to } t = T$$

$$V = L \frac{di}{dt} = L \frac{d}{dt} \left( \frac{-2I_0 t}{T} \right) = -\text{constant}$$

Therefore, the graph of option (D) is correct.

**Q18 Text Solution:**

(C)

From phasor diagram it is clear that current is lagging with respect to  $E_{\text{rms}}$ . This may be happen in LCR or LR circuit

**Q19 Text Solution:**

(D)

$$M = \frac{\mu_0 N_1 N_2 A}{l}$$

$$= \frac{4\pi \times 10^{-7} \times 300 \times 400 \times 10 \times 10^{-4}}{20 \times 10^{-2}}$$

$$= 24\pi \times 10^{-5} \text{ H}$$

$$= 2.4\pi \times 10^{-4} \text{ H}$$

**Q20 Text Solution:**

(D)

To cover the charge we need 12 geometry,

$$\phi = \frac{q}{12\epsilon_0}$$

**Q21 Text Solution:**

(C)

$$E = \frac{kqx}{(x^2 + R^2)^{3/2}}$$

**Q22 Text Solution:**

(B)



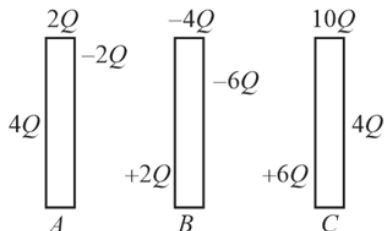
If potential of both sphere are same then there will be No flow of charge

$$\text{or } V_1 = V_2$$

$$\text{or } K \frac{2q}{a} = K \frac{3q}{b}$$

**Q23 Text Solution:**

(B)



$$Q_{net} = Q_{sys} = 2Q + (-4Q) + 10Q = 8Q$$

**Q24 Text Solution:**

(A)

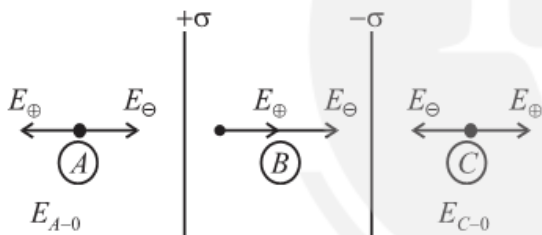
$$E_{net} = \sqrt{2}E, \text{ where } E = \frac{\sqrt{2}k\lambda}{R}$$

$$E_{net} = \frac{2k\lambda}{R}$$

$$\vec{E}_{net} = \frac{2k\lambda}{R} (-\hat{i})$$

**Q25 Text Solution:**

(B)



$$E_A = 0$$

$$E_B = \frac{\sigma}{2\epsilon_0} + \frac{\sigma}{2\epsilon_0}$$

$$E_B = \frac{2\sigma}{2\epsilon_0}$$

$$E_B = \frac{\sigma}{\epsilon_0}$$

**Q26 Text Solution:**

(A)

$$Q_{system} = 18 + 22 = 40 \text{ C}$$

$$\frac{Q'_1}{Q'_2} = \frac{10}{30} = \frac{1}{3}$$

$$\Rightarrow Q'_1 = 10 \text{ C} \quad Q'_2 = 30 \text{ C}$$

So, +8 C charge will flow from A to B.

**Q27 Text Solution:**

(D)

$$W_1 = -e[70\text{V} - 100\text{V}] = +30\text{eV}$$

$$W_2 = -e[70\text{V} - 90\text{V}] = +20\text{eV}$$

$$W_3 = -e[70\text{V} - 80\text{V}] = +10\text{eV}$$

$$W_4 = -e[80\text{V} - 100\text{V}] = +20\text{eV}$$

**Q28 Text Solution:**

(B)

(a) Initially inductor behaves like open circuit so current in  $4\Omega$  is  $= \frac{10}{2+4} = \frac{10}{6} \text{ A}$

(b) After some time inductor behaves like short circuit so current in  $4\Omega$  is zero.

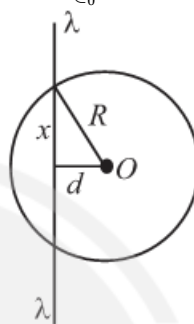
**Q29 Text Solution:**

(B)

$$\phi = \frac{2\lambda \sqrt{R^2 - d^2}}{\epsilon_0}$$

OR

$$\phi = \frac{\Sigma q}{\epsilon_0}$$



$\Sigma q$  = charge enclosed by Gaussian surface.

$\Sigma q = \lambda(\text{length of wire enclosed})$

$$= \lambda (2\sqrt{R^2 - d^2}) \Rightarrow \phi = \frac{2\lambda \sqrt{R^2 - d^2}}{\epsilon_0}$$

**Q30 Text Solution:**

(A)

$$E_{eq} = (10 + 5) = 15 \text{ V and}$$

$$r_{eq} = (3 + 2) = 5 \Omega$$

For maximum power in  $R$

$$R = r_{eq} = 5 \Omega$$

So current

$$I = \frac{E_{eq}}{R + r_{eq}} = \frac{15}{5+5} = 1.5 \text{ A}$$

Power in resistance  $R$

$$P = I^2 R = (1.5)^2 \times 5 \text{ W} = 11.25 \text{ W}$$

**Q31 Text Solution:**

(A)

Apply superposition of electric field using the fact that electric field is radially outwards.

**Q32 Text Solution:**

(C)

If middle shell and outer shell are connected by wire then

$$V_2 = V_3$$

If charge on middle shell is  $q$ .

$$\text{Charge on outer shell} = 3Q - 2Q - q$$

$$= Q - q$$



$$V_2 = \frac{kQ}{2R} + \frac{kq}{2R} + \frac{k(Q-q)}{3R}$$

$$V_3 = \frac{kQ}{3R} + \frac{kq}{3R} + \frac{k(Q-q)}{3R}$$

$$\Rightarrow \text{If } V_2 = V_3 \Rightarrow \frac{kQ}{2R} + \frac{kq}{2R} = \frac{kQ}{3R} + \frac{kq}{3R}$$

$$\Rightarrow \frac{q}{2} - \frac{q}{3} = \frac{Q}{3} - \frac{Q}{2} \Rightarrow q = -Q$$

**Q33 Text Solution:**

(D)

$$\vec{F} = i \vec{L} \times \vec{B}$$

$$L = \sqrt{(a-0)^2 + (a-0)^2} = \sqrt{2}a$$

**Q34 Text Solution:**

(D)

$$i l B = mg; \quad i = \frac{mg}{lB}$$

**Q35 Text Solution:**

(B)

$$\langle u_E \rangle = \langle u_B \rangle$$

**Q36 Text Solution:**

(A)

$c_2$  and  $c_3$  are connected in series –

Equivalent capacitance

$$c' = \frac{1}{400} + \frac{1}{400} = 200 \mu F$$

Now  $c$  and  $c'$  are in parallel. Equivalent capacitance

$$c'' = c_1 + c' = 200 + 100 = 300 \mu F$$

Now  $c''$  and  $c_4$  are connected in series –

$$\frac{1}{c_{eq}} = \frac{1}{c_4} + \frac{1}{c''} = \frac{1}{300} + \frac{1}{400}$$

$$c_{eq} = \frac{1200}{7} \mu F$$

**Q37 Text Solution:**

(C)

$$B = \frac{\mu_0 \mu_r N i}{2\pi R}$$

$$= \frac{4\pi \times 10^{-7} \times 1000 \times 6000 \times 1}{2\pi \times 10 \times 10^{-2}} = 12 \text{ T}$$

**Q38 Text Solution:**

(D)

$$\mu = NIA$$

$$B = \frac{N\mu_0 I}{2R}$$

$$\Rightarrow I = \frac{2BR}{N\mu_0}$$

$$\mu = \frac{2BA}{\mu_0} \sqrt{\frac{A}{\pi}}$$

**Q39 Text Solution:**

(A)

We know that, flux associated with ring

$$\phi = NBA$$

$$\text{But } \phi = LI$$

$$\therefore LI = NBA$$

$$L = \frac{NBA}{l}$$

$$= \frac{N \cdot \frac{N\mu_0 I}{2a} \cdot A}{l} \left[ \because B = \frac{N\mu_0 I}{2a} \right]$$

$$= N^2 \cdot \frac{\mu_0}{2a} A = \left( \frac{1}{2\pi a} \right)^2 \cdot \frac{\mu_0}{2a}$$

$$\cdot \pi a^2 \left[ \because N = \frac{l}{2\pi a} \right]$$

$$= \mu_0 l^2 / 8a\pi$$

$$\therefore L \propto a^{-1}$$

**Q40 Text Solution:**

(B)

We know that quality factor ( $Q$ ),

$$Q = \frac{1}{R} \sqrt{\frac{L}{C}}$$

$$Q^2 = \frac{1}{R^2 C} \quad \text{or} \quad C = \frac{L}{Q^2 R^2}$$

**Q41 Text Solution:**

(B)

$$\text{Self-inductance, } L = \frac{\mu_0 N^2}{l} A$$

Here,  $N$  = total number of turns

When  $N$  and  $l$  are doubled, then

$$L' = \mu_0 \frac{(2N)^2}{2l} A = 2\mu_0 \frac{N^2 A}{l} = 2L$$

**Q42 Text Solution:**

(D)

Let the velocity of wire is  $v$  at time  $t$  and the emf induced is  $\varepsilon = vBl$ , then charge

(in magnitude) appeared on the plates of

capacitor is

$$q = C\varepsilon = C v B l$$

$$\text{Now current, } I = \frac{dq}{dt} = C B l \frac{dv}{dt} = C B l a$$

$$\left( \because a = \frac{dv}{dt} \right)$$

The force due to the current is

$$F' = IlB = C B^2 l^2 a$$

So, net force in the direction of motion of conductor wire

$$F - F' = ma \Rightarrow F = ma + C B^2 l^2 a$$

$$a = \frac{F}{m + C B^2 l^2}$$

**Q43 Text Solution:**

(C)

The torque on a solenoid of turns is

$$\tau = NIBA \sin \theta$$

$$= 2000 \times 2 \times 5 \times 10^{-2} \times 1.5 \times 10^{-4}$$

$$\times \sin 30^\circ$$

$$= 1.5 \times 10^{-2} \text{ N-m}$$

**Q44 Text Solution:**

(B)

Then find refractive index





$$\mu = \frac{c}{v}, v = \frac{\omega}{k} = \frac{6 \times 10^8}{4} = 1.5 \times 10^8$$

$$\mu = \frac{3 \times 10^8}{1.5 \times 10^8} = 2$$

**Q45 Text Solution:****(D)**

$$\text{Time constant, } \tau = \frac{L}{R} = \frac{4}{2} = 2$$

Expression of current in circuit is given by

$$I = \frac{e}{R} (1 - e^{-t/\tau}) = \frac{1}{2} (1 - e^{-2/2})$$

$$= \frac{1}{2} (1 - e^{-1}) \text{ A}$$

**Q46 Text Solution:****(B)**

In RC circuit,

$$\text{impedance, } Z = \sqrt{R^2 + X_C^2}$$

$$= \sqrt{R^2 + \left(\frac{1}{\omega C}\right)^2}$$

On placing dielectric in between the plates of capacitor, C will increase

i.e. Z will decrease and current will increase

$$I_R^B > I_R^A$$

**Q47 Text Solution:****(A)**

The charge in a DC circuit is

$$q = q_0 e^{-t/RC} = CV_0 e^{-t/RC} (\because q_0 = CV_0)$$

$$\text{The potential, } V_C = \frac{q}{C} = V_0 e^{-t/RC}$$

Higher value of R i.e.,  $e^{t/RC}$  is smaller i.e.,  $V_C$  is higher

$$V_3 > V_2 > V_1$$

$$3 \rightarrow A, 2 \rightarrow B, 1 \rightarrow C$$

**Q48 Text Solution:****(A)**The phase angle between voltage V and current I is  $\frac{\pi}{2}$ .

Power consumer,

$$P = V_{\text{rms}} \times I_{\text{rms}} \times \cos \phi$$

$$P = V_{\text{rms}} \times I_{\text{rms}} \times \cos \frac{\pi}{2} = 0 (\because \phi = \frac{\pi}{2})$$

$$\cos 90^\circ = 0$$

$$\therefore P = 0$$

**Q49 Text Solution:****(C)**Here, inductor also have internal resistance ( $r_L$ ).So, total resistance ( $R$ ) =  $40 + 40 = 80 \Omega$ The impedance of  $L - C - R$  circuit is given by

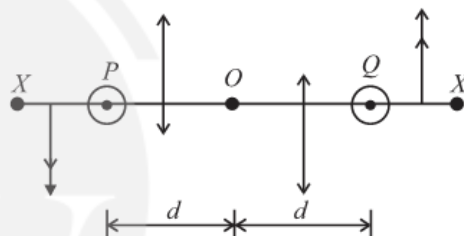
$$Z = \sqrt{R^2 + (X_L - X_C)^2}$$

$$= \sqrt{(80)^2 + (100 - 40)^2} = 100 \Omega$$

**Q50 Text Solution:****(B)**

The steady currents in the two parallel wires flow out of paper. O is a point in the centre between two wires. The wires are at P and Q. Since the current flows out of paper, the magnetic field at points to the right of given wire will be upwards and to the left will be downwards.

(i) At point O, net field will be zero. The fields due to wires at P and Q are equal in magnitude but opposite in directions.



(ii) Region QX': Magnetic field will be upwards. The region lies to right of both the wires.

(iii) Region OQ: Net field will be downwards. Due to nearness, wire at Q dominates.

(iv) Region OP: Net field will be upwards. Due to nearness, wire at P dominates.

(v) Region PX: Net field will be downwards. The region lies to left of both the wires.

Graph (b) satisfies all the five conditions.

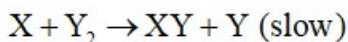
**Q51 Text Solution:****(B)**

For reversible adiabatic expansion

$$T_1 V_1^{\gamma-1} = T_2 V_2^{\gamma-1} \Rightarrow \frac{T_2}{T_1} = \left(\frac{V_1}{V_2}\right)^{\gamma-1} = \left(\frac{1}{8}\right)^{5/3-1}$$

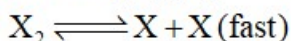
$$\text{Hence, } T_2 : T_1 = 1 : 4$$

**Q52 Text Solution:****(D)**



As rate depends upon the slowest step,

$$\text{Rate} = k [X] [Y_2] \dots\dots(i)$$



$$K_{eq} = \frac{[X]^2}{[X_2]} \text{ or } [X] = K_{eq}^{1/2} [X_2]^{1/2}$$

Substituting in (i),

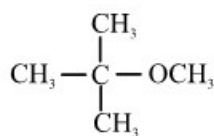
$$\text{Rate} = k K_{eq}^{1/2} [X_2]^{1/2} [Y_2] = k' [X_2]^{1/2} [Y_2]$$

$$\text{Overall order} = \frac{1}{2} + 1 = \frac{3}{2} = 1.5$$

**Q53 Text Solution:**

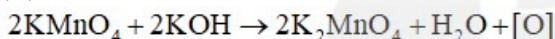
(C)

2-Methoxy-2-methylpropane is:



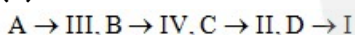
**Q54 Text Solution:**

(A)



**Q55 Text Solution:**

(D)



**Q56 Text Solution:**

(B)

$\text{HOOC}-\text{CH}_2-\text{COOH}$ : Malonic acid

**Q57 Text Solution:**

(D)

The small increase in ionization enthalpy from Sn to Pb is due to a considerable increase in nuclear charge which outweighs the shielding effect of all the electrons in the inner shells including those of 4f and 5d – electrons and inert pair effect.

**Q58 Text Solution:**

(A)

Large difference in boiling points of alcohols and ethers is due to the presence of hydrogen bonding in alcohols and oxygen of ether can also form hydrogen bonding with water molecules.

**Q59 Text Solution:**

(D)

Both (2) & (3) are correct.

**Q60 Text Solution:**

(B)

The factor  $e^{-E_a/RT}$  corresponds to the fraction of molecules that have kinetic energy greater than  $E_a$ .

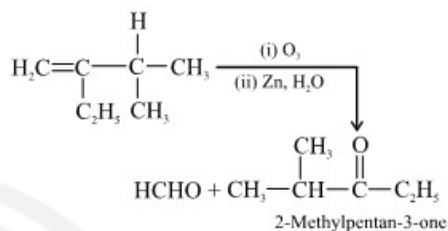
**Q61 Text Solution:**

(A)

The mutual screening offered in case of 5f-electrons is less than that in the 4f-electrons.

**Q62 Text Solution:**

(A)



**Q63 Text Solution:**

(D)

Hybridisation of  $\text{XeF}_2$  is  $\text{sp}^3\text{d}$  having linear shape.



**Q64 Text Solution:**

(C)

Heat of neutralization of HCN

= Total heat – heat of neutralization of strong base

$$= -2.460 - (-13.71) = 11.25 \text{ kcal or } 11250 \text{ cal.}$$

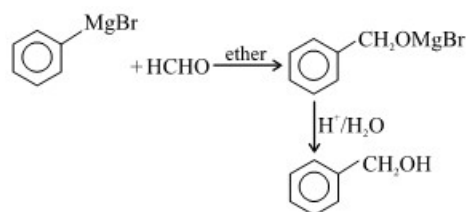
**Q65 Text Solution:**

(C)

Hydrolysis of cane sugar is pseudo first order reaction. Since, water is always in excess, rate of reaction does not depend appreciably on its concentration, thus it is an example of pseudo first order reactions.

**Q66 Text Solution:**

(D)



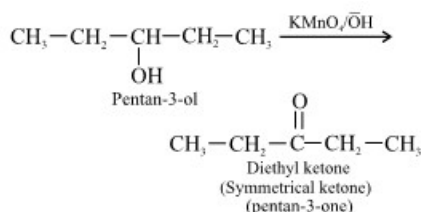
**Q67 Text Solution:**

(C)

$\text{MnO}_2$  and  $\text{Cr}_2\text{O}_3$  are amphoteric oxides

**Q68 Text Solution:**

(4)



**Q69 Text Solution:**

(D)

$\Delta H^\circ$  and  $\Delta G^\circ$  remains constant in the given temperature range.

$$\Delta G^\circ = \Delta H^\circ - T\Delta S^\circ$$

$$\Delta S^\circ = \frac{\Delta H^\circ - \Delta G^\circ}{T} = \frac{41.8 + (-66.9)}{300}$$

$$= -\frac{25.1}{300} = -0.08367 \text{ kJ K}^{-1} \text{ mol}^{-1}$$

$\therefore$

$$\Delta G^\circ = \Delta H^\circ - T\Delta S^\circ = ((-66.9) - 330(-0.0836))$$

$$= -39.31 \text{ kJ mol}^{-1}$$

**Q70 Text Solution:**

(D)

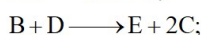
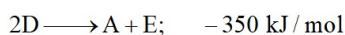
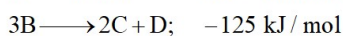
$\text{Ti}^{3+}$  has  $[\text{Ar}]3d^1$  configuration. Thus, d-d transition is possible and thereby it shows colour.

**Q71 Text Solution:**

(B)

Adding all the equations.

$\Delta H$



$$\Delta H = (300 - 125 - 350) = -175 \text{ kJ/mol}$$

**Q72 Text Solution:**

(2)

$\beta$ -Black phosphorus does not burn in air upto 673K.

**Q73 Text Solution:**

(D)

Chemistry of actinoids is more complicated due to the possibility of large number of oxidation states.

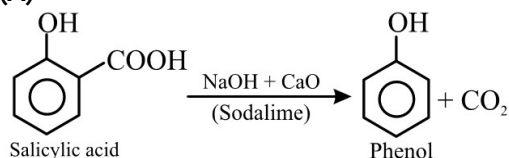
**Q74 Text Solution:**

(B)

Both the allotropes of sulphur are stable and at equilibrium at a temperature of 369 K and this temperature is known as transition temperature. In other words, we can say that  $\alpha$ -Sulphur is stable below 369 K and it becomes  $\beta$ -Sulphur above that temperature.

**Q75 Text Solution:**

(A)



**Q76 Text Solution:**

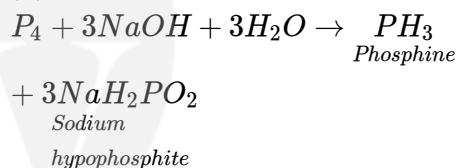
(D)

$$\% \text{ of } B = \frac{k_1}{k_1 + k_2} = \frac{1.26 \times 10^{-4}}{1.26 \times 10^{-4} + 3.8 \times 10^{-5}} \times 100$$

$$\frac{1.26 \times 10^{-4}}{10^{-4}(1.26 + 0.38)} \times 100 = 76.38\%$$

**Q77 Text Solution:**

(C)



**Q78 Text Solution:**

(B)

$$\text{For first order reaction, } t = \frac{2.303}{k} \log \frac{a}{a-x}$$

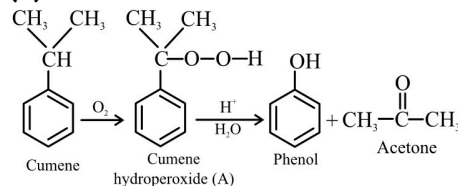
$$t_{99.9\%} = \frac{2.303}{k} \log \frac{100}{100-99.9} = \frac{2.303}{k} \times 3$$

$$t_{50\%} = \frac{2.303}{k} \log \frac{100}{50} = \frac{2.303}{k} \log 2 = \frac{2.303}{k} \times 0.301$$

$$\frac{t_{99.9\%}}{t_{50\%}} = 10 (\text{approx}).$$

**Q79 Text Solution:**

(C)



**Q80 Text Solution:**

(C)

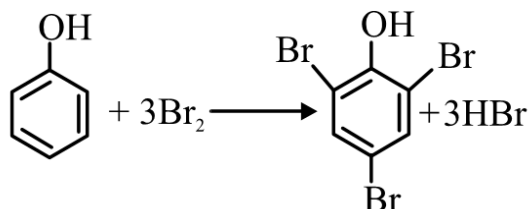




$$\begin{aligned}
 &= \frac{2.303}{t} \log \left[ \frac{P_0}{P_0 - \left( \frac{P_t - P_0}{2} \right)} \right] \\
 &= \frac{2.303}{t} \log \left( \frac{2P_0}{2P_0 - P_t + P_0} \right) \\
 &= \frac{2.303}{t} \log \left( \frac{2P_0}{3P_0 - P_t} \right)
 \end{aligned}$$

**Q92 Text Solution:**

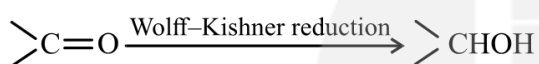
(C)



0.24 mole of phenol reacts with  $3 \times 0.24 = 0.72$  mole of  $\text{Br}_2$ .

**Q93 Text Solution:**

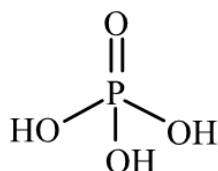
(B)



**Q94 Text Solution:**

(C)

Orthophosphoric acid is  $\text{H}_3\text{PO}_4$ .

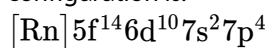


Orthophosphoric acid

**Q95 Text Solution:**

(B)

Its atomic number is 116 hence electronic configuration is.



**Q96 Text Solution:**

(D)

$A \rightarrow B$  (uncatalysed reaction)

$A \rightarrow B$  (catalysed reaction)

$$k_{\text{uncat.}} = A e^{-E_{a(\text{uncat.})}/RT}$$

$$k_{\text{cat.}} = A e^{-E_{a(\text{cat.})}/RT}$$

$$\frac{k_{\text{cat.}}}{k_{\text{uncat.}}} = e^{(E_{a(\text{uncat.})} - E_{a(\text{cat.})})/RT \times \frac{1}{RT}}$$

$$\frac{k_{\text{cat.}}}{k_{\text{uncat.}}} = e^{\frac{8.314 \times 10^3}{8.314 \times 300}} = e^{3.33} = 28 \text{ times}$$

**Q97 Text Solution:**

(D)

$\alpha$ -Halogenation of carboxylic acid is called Hell-Volhard-Zelinsky (HVZ) reaction.

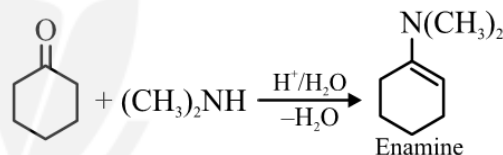
**Q98 Text Solution:**

(3)

Due to lanthanide contraction, the size of  $\text{Lu}^{3+}$  is decreased and therefore  $\text{Lu}(\text{OH})_3$  shows more covalent character (small cation, more is covalent character - Fajan's rule). Thus  $\text{La}(\text{OH})_3$  is more ionic and thus more basic.

**Q99 Text Solution:**

(B)



**Q100 Text Solution:**

(3)

Aldehydes and ketones with  $\alpha$ -hydrogen atom can undergo Aldol condensation while without  $\alpha$ -hydrogen atom can undergo Cannizzaro reaction.

**Q101 Text Solution:**

(A)

All solids that settle form the primary sludge during primary treatment.

[New NCERT Class 12<sup>th</sup> Page No. 154]

**Q102 Text Solution:**

(C)

- Inflorescences are formed which bear the floral buds and then the flowers.
- Much before the actual flower is seen on a plant, the decision that the plant is going to flower has taken place.



- Androecium consists of a whorl of stamens
- Gynoecium represents the female reproductive organ.

[New NCERT Class 12<sup>th</sup> Page No. 4]

**Q103 Text Solution:**

(D)

Queen Victoria of England was a carrier for haemophilia. Since this is an X-linked disease, it can be predicted that haemophilia would have occurred in more of her male than her female descendants.

[New NCERT Class 12<sup>th</sup> Page No. 74]

**Q104 Text Solution:**

(D)

Morgan carried out several dihybrid crosses in *Drosophila* to study genes that were sex-linked.

[New NCERT Class 12<sup>th</sup> Page No. 67]

**Q105 Text Solution:**

(C)

Phenotypic and genotypic ratio is equal in TtRr × ttrr. It represents a test cross.

[New NCERT Class 12<sup>th</sup> Page No. 58]

**Q106 Text Solution:**

(C)

|              |             |
|--------------|-------------|
| Co-dominance | Blood group |
|--------------|-------------|

[New NCERT Class 12<sup>th</sup> Page No. 61]

**Q107 Text Solution:**

(D)

Embryo develops at the micropylar end of the embryo sac where the zygote is situated. Most zygotes divide only after certain amount of endosperm is formed. This is an adaptation to provide assured nutrition to the developing embryo. Though the seeds differ greatly, the early stages of embryo development (embryogeny) are similar in both monocotyledons and dicotyledons.

[New NCERT Class 12<sup>th</sup> Page No. 19]

**Q108 Text Solution:**

(C)

If the female parent bears bisexual flowers, removal of anthers from the flower bud before the anther dehisces using a pair of forceps is necessary. This step is referred to as emasculation. Emasculated flowers have to be

covered with a bag of suitable size, generally made up of butter paper, to prevent contamination of its stigma with unwanted pollen. If the female parent produces unisexual flowers, there is no need for emasculation. The female flower buds are bagged before the flowers open.

[New NCERT Class 12<sup>th</sup> Page No. 17]

**Q109 Text Solution:**

(B)

|                                            |     |
|--------------------------------------------|-----|
| Number of embryo sac in an ovule generally | One |
|--------------------------------------------|-----|

[New NCERT Class 12<sup>th</sup> Page No. 10]

**Q110 Text Solution:**

(D)

|                               |                        |
|-------------------------------|------------------------|
| Incomplete dominance          | <i>Antirrhinum</i> sp. |
| Linkage                       | <i>Drosophila</i>      |
| Sex linked recessive disorder | Colour blindness       |
| Autosomal recessive disorder  | Sickle cell anaemia    |

[New NCERT Class 12<sup>th</sup> Page No. 60, 67, 73, 74]

**Q111 Text Solution:**

(D)

The genes white and yellow were very tightly linked and showed only 1.3 per cent recombination while white and miniature wing showed 37.2 per cent recombination.

[New NCERT Class 12<sup>th</sup> Page No. 67]

**Q112 Text Solution:**

(B)

Failure of segregation of chromatids during cell division cycle results in the gain or loss of a chromosome(s), called aneuploidy. For example, Down's syndrome results in the gain of extra copy of chromosome 21.

[New NCERT Class 12<sup>th</sup> Page No. 75]

**Q113 Text Solution:**

(A)

'Toddy', a traditional drink of some parts of southern India is made by fermenting sap from palms.

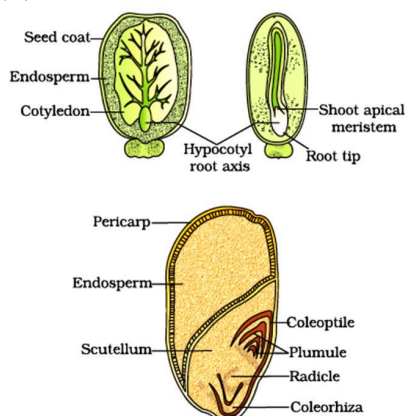
[New NCERT Class 12<sup>th</sup> Page No. 151]





**Q114 Text Solution:**

(B)



| I                   | II                    | III       | IV        |
|---------------------|-----------------------|-----------|-----------|
| Hypocotyl root axis | Shoot apical meristem | Scutellum | Endosperm |

[New NCERT Class 12<sup>th</sup> Page No. 19]

**Q115 Text Solution:**

(D)

Pollen grain exine has prominent apertures called germ pores where sporopollenin is absent.

[New NCERT Class 12<sup>th</sup> Page No. 7]

**Q116 Text Solution:**

(A)

The bilobed nature of an anther is very distinct in the transverse section of the anther.

[New NCERT Class 12<sup>th</sup> Page No. 5]

**Q117 Text Solution:**

(A)

$2n \rightarrow \text{Zygote} \rightarrow \text{Syngamy}$

$3n \rightarrow \text{PEC} / \text{PEN} \rightarrow \text{Triple fusion}$

[New NCERT Class 12<sup>th</sup> Page No. 18]

**Q118 Text Solution:**

(B)

Mendel proposed a second set of generalisations that we call Mendel's Law of Independent Assortment. The law states that 'when two pairs of traits are combined in a hybrid, segregation of one pair of characters is independent of the other pair of characters'.

[New NCERT Class 12<sup>th</sup> Page No. 64]

**Q119 Text Solution:**

(D)

In 1900, three Scientists (de Vries, Correns and von Tschermak) independently rediscovered Mendel's results on the inheritance of characters.

[New NCERT Class 12<sup>th</sup> Page No. 65]

**Q120 Text Solution:**

(D)

- Majority of insect-pollinated flowers have large, colourful, fragrant and rich in nectar.
- Pollination does not guarantee the transfer of the right type of pollen.

[New NCERT Class 12<sup>th</sup> Page No. 14, 15]

**Q121 Text Solution:**

(A)

The two-celled stage of mature pollen grain consists of vegetative cell and generative cell.

[New NCERT Class 12<sup>th</sup> Page No. 7]

**Q122 Text Solution:**

(D)

Mendelian disorders are mainly determined by alteration or mutation in the single gene. Colour blindness occurs in about 8 per cent of males and only about 0.4 per cent of females.

[New NCERT Class 12<sup>th</sup> Page No. 73]

**Q123 Text Solution:**

(D)

Activated sludge is sediment present in settlement tank of sewage treatment plant which is a right source of aerobic bacteria.

[New NCERT Class 12<sup>th</sup> Page No. 154]

**Q124 Text Solution:**

(C)

The large holes in 'Swiss cheese' are due to production of a large amount of  $\text{CO}_2$  by a bacterium named *Propionibacterium sharmanii*.

[New NCERT Class 12<sup>th</sup> Page No. 151]

**Q125 Text Solution:**

(D)

*Clostridium butylicum* - Butyric acid

[New NCERT Class 12<sup>th</sup> Page No. 153]

**Q126 Text Solution:**

(A)

- Micro-organisms such as *Lactobacillus* and others commonly called lactic acid bacteria (LAB) grow in milk and convert it to curd.



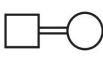
During growth, the LAB produce acids that coagulate and partially digest the milk proteins. improves its nutritional quality by increasing vitamin B<sub>12</sub>. In our stomach too, the LAB play very beneficial role in checking disease causing microbes.

- Bt-cotton is being cultivated in some states of our country.
- *Nucleopolyhedrovirus* do not show effect on birds and fishes.

[New NCERT Class 12<sup>th</sup> Page No. 151, 157]

**Q127 Text Solution:**

(D)

|                                                                                     |                           |
|-------------------------------------------------------------------------------------|---------------------------|
|    | Consanguineous mating     |
|    | Sex unspecified           |
|    | Five unaffected offspring |
|  | Normal individuals        |

[New NCERT Class 12<sup>th</sup> Page No. 72]

**Q128 Text Solution:**

(A)

- Penicillin was extensively used to treat American soldiers wounded in World War II.
- Fleming, Chain and Florey were awarded the Nobel Prize in 1945 for the discovery of penicillin.
- Full potential as an effective antibiotic was established much later by Ernest Chain and Howard Florey.

[New NCERT Class 12<sup>th</sup> Page No. 152]

**Q129 Text Solution:**

(A)

A biological control being developed for use in the treatment of plant disease is the fungus *Trichoderma*. *Trichoderma* species are free-living fungi that are very common in the root ecosystems. They are effective biocontrol agents of several plant pathogens.

[New NCERT Class 12<sup>th</sup> Page No. 157]

**Q130 Text Solution:**

(C)

Streptokinase produced by the bacterium *Streptococcus* and modified by genetic engineering is used as a 'clot buster' for removing clots from the blood vessels of patients who have undergone myocardial infarction leading to heart attack.

[New NCERT Class 12<sup>th</sup> Page No. 153]

**Q131 Text Solution:**

(C)

The 'Roquefort cheese' are ripened by growing a specific fungus on them, which gives them a particular flavour.

[New NCERT Class 12<sup>th</sup> Page No. 151]

**Q132 Text Solution:**

(A)

Water pollinated plants are *Vallisneria* and *Hydrilla* which grow in fresh water and several marine sea-grasses such as *Zostera*. Not all aquatic plants use water for pollination. In a majority of aquatic plants such as water hyacinth and water lily, the flowers emerge above the level of water and are pollinated by insects or wind as in most of the land plants. In seagrasses, female flowers remain submerged in water and the pollen grains are released inside the water.

[New NCERT Class 12<sup>th</sup> Page No. 11]

**Q133 Text Solution:**

(B)

Mendel's experiments had large sample size which gave greater credibility to the data.

[New NCERT Class 12<sup>th</sup> Page No. 54]

**Q134 Text Solution:**

(C)

Wine and beer are produced without distillation whereas whisky, brandy and rum are produced by distillation of the fermented broth.

[New NCERT Class 12<sup>th</sup> Page No. 152]

**Q135 Text Solution:**

(B)

| Genotype of offspring | Blood types of offspring |
|-----------------------|--------------------------|
| $I^A I^A$             | A                        |
| $I^A i$               | A                        |





|           |    |
|-----------|----|
| $I^A I^B$ | AB |
| $I^B i$   | B  |
| $ii$      | O  |

[New NCERT Class 12<sup>th</sup> Page No. 61]

**Q136 Text Solution:**

(C)

Bacteria can fix atmospheric nitrogen while free-living in the soil (examples *Azospirillum* and *Azotobacter*), thus enriching the nitrogen content of the soil.

[New NCERT Class 12<sup>th</sup> Page No. 158]

**Q137 Text Solution:**

(A)

Mendel's investigations into inheritance patterns it was for the first time that statistical analysis and mathematical logic were applied to problems in biology.

[New NCERT Class 12<sup>th</sup> Page No. 54]

**Q138 Text Solution:**

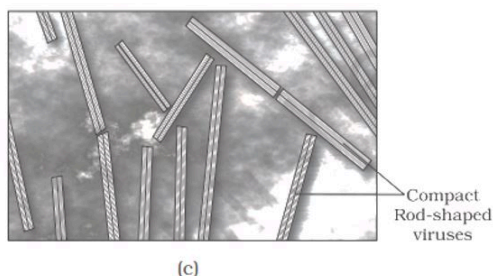
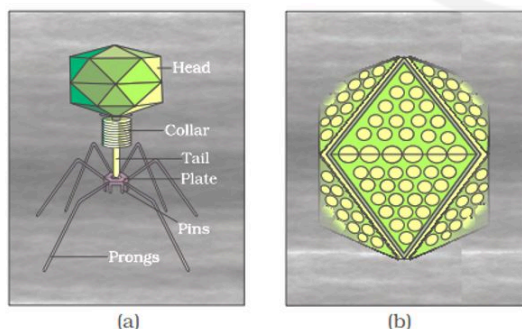
(D)

|               |                         |
|---------------|-------------------------|
| Cyclosporin-A | Immunosuppressive agent |
| Proteases     | Clarifying bottles      |
| Streptokinase | Myocardial infarction   |
| Lipases       | Oily stains             |

[New NCERT Class 12<sup>th</sup> Page No. 153]

**Q139 Text Solution:**

(A)



a-Bacteriophage; b-Adenovirus; c-Tobacco Mosaic Virus (TMV)

[New NCERT Class 12<sup>th</sup> Page No. 150]

**Q140 Text Solution:**

(C)

The secondary treatment of sewage waste is also known as biological treatment.

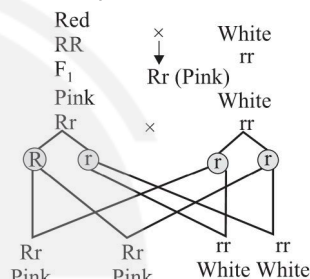
[New NCERT Class 12<sup>th</sup> Page No. 154]

**Q141 Text Solution:**

(A)

In dog flower, red flowered plant crossed with white flowered one resulted in all pink flowered plants.

But there are zero percent chances of appearance of red flowered plants in a cross between pink and white flowered plant.



[New NCERT Class 12<sup>th</sup> Page No. 60]

**Q142 Text Solution:**

(D)

Person affected from Turner's syndrome has total 45 chromosomes in each cell.

[New NCERT Class 12<sup>th</sup> Page No. 76]

**Q143 Text Solution:**

(D)

In some cereals such as rice and wheat, pollen grains lose viability within 30 minutes of their release and in some members of Rosaceae, Leguminosae and Solanaceae, they maintain viability for months.

[New NCERT Class 12<sup>th</sup> Page No. 8]

**Q144 Text Solution:**

(B)

Flowering plants have developed many devices to discourage self pollination and to encourage cross-pollination.

[New NCERT Class 12<sup>th</sup> Page No. 15]

**Q145 Text Solution:**

(A)



Dragonflies are useful to get rid of mosquitoes.

[New NCERT Class 12<sup>th</sup> Page No. 151]

**Q146 Text Solution:**

(A)

$\alpha$  thalassemia and  $\beta$  thalassemia in humans is controlled by genes on chromosome 16 and 11 respectively.

[New NCERT Class 12<sup>th</sup> Page No. 75]

**Q147 Text Solution:**

(C)

Biocontrol refers to the use of biological methods for controlling plant diseases and pests.

In agriculture, there is a method of controlling pests that relies on natural predation rather than introduced chemicals.

[New NCERT Class 12<sup>th</sup> Page No. 156]

**Q148 Text Solution:**

(D)

*Bacillus thuringiensis* (often written as Bt). These are available in sachets as dried spores which are mixed with water and sprayed onto vulnerable plants such as brassicas and fruit trees, where these are eaten by the insect larvae. In the gut of the larvae, the toxin is released and the larvae get killed. The bacterial disease will kill the caterpillars, but leave other insects unharmed.

[New NCERT Class 12<sup>th</sup> Page No. 157]

**Q149 Text Solution:**

(C)

In Down Syndrome the affected individual is short statured with small round head, furrowed tongue and partially open mouth. Palm is broad with characteristic palm crease. Physical, psychomotor and mental development is retarded.

[New NCERT Class 12<sup>th</sup> Page No. 76]

**Q150 Text Solution:**

(B)

Pollen grains are generally spherical measuring about 25-50 micrometers in diameter.

[New NCERT Class 12<sup>th</sup> Page No. 7]

**Q151 Text Solution:**

(C)

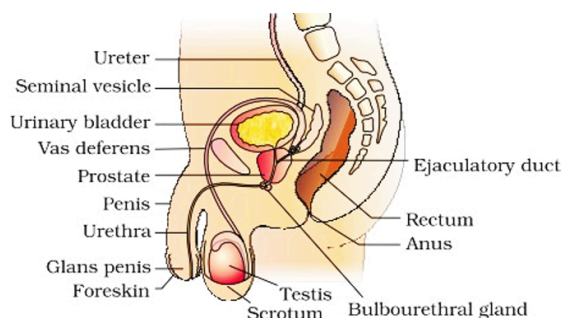
In disruptive selection, more individuals acquire peripheral character value at both ends of the

distribution curve.

[New NCERT Class 12<sup>th</sup> Page No. 120]

**Q152 Text Solution:**

(A)



- Ejaculatory duct represents common duct which forms by the fusion of duct of seminal vesicle and vas deferens.

[New NCERT Class 12<sup>th</sup> Page No. 27]

**Q153 Text Solution:**

(A)

- Lactational amenorrhea is a natural method of contraception.
- As no medicines or devices are used in these methods, so side effects are almost nil. Chances of failure, though, of this method are also high.

[New NCERT Class 12<sup>th</sup> Page No. 44]

**Q154 Text Solution:**

(B)

Secondary polar body is released in the oviduct after the completion of meiotic-II division.

[New NCERT Class 12<sup>th</sup> Page No. 49, 50]

**Q155 Text Solution:**

(C)

- Placental mammals: Lemur
- Australian marsupials: Spotted cuscus

[New NCERT Class 12<sup>th</sup> Page No. 118]

**Q156 Text Solution:**

(B)

- The edges of the infundibulum possess finger-like projections called fimbriae.



- Fimbriae help in collection of the ovum after ovulation.

[New NCERT Class 12<sup>th</sup> Page No. 29]

**Q157 Text Solution:**  
(C)

- Menstrual cycle on average is 28 day long and ovulation occurs at 14<sup>th</sup>
- Regardless of the length of menstrual cycle, luteal phase is 14 day long.
- In a 40-day menstrual cycle, ovulation is expected to happen on the 26<sup>th</sup> day as,  $40 - 14 = 26^{\text{th}}$

[New NCERT Class 12<sup>th</sup> Page No. 33,34]

**Q158 Text Solution:**  
(A)

Oparin of Russia and Haldane of England proposed that the first form of life could have come from pre-existing non-living organic molecules.

[New NCERT Class 12<sup>th</sup> Page No. 111]

**Q159 Text Solution:**  
(B)

Antrum is a fluid filled cavity present in the tertiary follicles of ovary.

[New NCERT Class 12<sup>th</sup> Page No. 32]

**Q160 Text Solution:**  
(B)

The earth is about 4000 years old according to the special creation theory.

[New NCERT Class 12<sup>th</sup> Page No. 112]

**Q161 Text Solution:**  
(D)

- Cervical caps are inserted into the female reproductive tract to cover the cervix during coitus.
- The hormone releasing IUDs make the uterus unsuitable for implantation.

[New NCERT Class 12<sup>th</sup> Page No. 44,45]

**Q162 Text Solution:**  
(C)

Mutations are random and directionless while Darwinian variations are small and directional.

[New NCERT Class 12<sup>th</sup> Page No. 119]

**Q163 Text Solution:**  
(C)

- The mitotic division starts as the zygote moves through the isthmus of the oviduct called cleavage towards the uterus called cleavage and forms 2, 4, 8, 16 daughter cells called blastomeres.
- The embryo with 8 to 16 blastomeres is called a morula. The morula continues to divide and transforms into blastocyst as it moves further into the uterus.

[New NCERT Class 12<sup>th</sup> Page No. 36]

**Q164 Text Solution:**  
(C)

- Maternal mortality rate and Infant mortality rate have decreased due to improved medical facilities. Death rate has fallen due to improved personal hygiene sanitation and modern medicines.
- If there is a decline in number of people in the reproducible age population explosion will not occur rather there will be a decrease in population.

[New NCERT Class 12<sup>th</sup> Page No. 43]

**Q165 Text Solution:**  
(B)

Hugo deVries believed mutation caused speciation and hence called it saltation which means single step large mutation.

[New NCERT Class 12<sup>th</sup> Page No. 119]

**Q166 Text Solution:**  
(B)

- LNG-20 is hormone releasing IUDs.
- Cervical caps and vaults are physical contraceptive barriers.
- Multiload 375 and Cu7 are copper releasing IUDs.
- Lactational amenorrhea is a natural contraceptive method.

[New NCERT Class 12<sup>th</sup> Page No.44]

**Q167 Text Solution:**  
(D)

Genital warts is a sexually transmitted disease.



[New NCERT Class 12<sup>th</sup> Page No. 47]**Q168 Text Solution:**

(C)

The female external genitalia include mons pubis, labia majora, labia minora, hymen and clitoris.

[New NCERT Class 12<sup>th</sup> Page No. 30]**Q169 Text Solution:**

(C)

Spermicidal creams, jellies and foams are usually used along with the barriers to increase their contraceptive efficiency.

[New NCERT Class 12<sup>th</sup> Page No. 44]**Q170 Text Solution:**

(D)

- Contraceptive pills are very effective with lesser side effects used by females.
- They work by inhibiting ovulation and implantation.
- They alter the quality of cervical mucus to prevent/retard the entry of sperms.

[New NCERT Class 12<sup>th</sup> Page No. 45]**Q171 Text Solution:**

(C)

- Fertilisation can only occur if the ovum and sperms are transported simultaneously to the ampullary region.
- This is the reason why not all copulations lead to fertilisation and pregnancy.

[New NCERT Class 12<sup>th</sup> Page No. 35]**Q172 Text Solution:**

(D)

The Neanderthal man with a brain size of 1400cc lived in near east and central Asia between 1,00,000-40,000 years back. They used hides to protect their body and buried their dead.

[New NCERT Class 12<sup>th</sup> Page No. 125]**Q173 Text Solution:**

(C)

|                              |                                 |
|------------------------------|---------------------------------|
| Reproductive tract infection | Chlamydia                       |
| Condom                       | Prevents sperms reaching cervix |
| Tubectomy                    | Sterilisation in females        |

Multiload375

Copper releasing IUDs

[New NCERT Class 12<sup>th</sup> Page No. 42,44,45]**Q174 Text Solution:**

(D)

The frequency of dominant allele( $p$ ) = 0.7

From Hardy Weinberg principle,  $p + q = 1$

Frequency of recessive allele( $q$ ) = 0.3

So, the frequency of heterozygotes =  $2pq$   
= 0.42

[New NCERT Class 12<sup>th</sup> Page No. 120]**Q175 Text Solution:**

(D)

- Vaults is a type of barrier method of contraception.
- It covers the cervix during coitus.
- It is made of rubber.
- It is reusable.

[New NCERT Class 12<sup>th</sup> Page No. 44]**Q176 Text Solution:**

(B)

Secondary oocyte is released from the ovary at the time of ovulation.

[New NCERT Class 12<sup>th</sup> Page No. 32]**Q177 Text Solution:**

(B)

Sweet potato (root modification) and potato (stem modification) is example for analogy.

[New NCERT Class 12<sup>th</sup> Page No. 115]**Q178 Text Solution:**

(D)

Periodic abstinence is a natural method of contraception, in which no device or chemical is used.

[New NCERT Class 12<sup>th</sup> Page No. 44]**Q179 Text Solution:**

(D)

Branching descent and natural selection are the two key concepts of Darwinian Theory of Evolution.

[New NCERT Class 12<sup>th</sup> Page No. 118]**Q180 Text Solution:**

(C)



- In ICSI (intra cytoplasmic sperm injection) procedure sperms are directly injected into the ovum to form an embryo.
- Infertility in male partner could be corrected by artificial insemination.

[New NCERT Class 12<sup>th</sup> Page No. 47]

**Q181 Text Solution:**

(B)

|                            |                          |
|----------------------------|--------------------------|
| Fish like reptiles         | <i>Ichthyosaurs</i>      |
| Dagger like teeth          | <i>Tyrannosaurus rex</i> |
| Pre-historic cave art      | 18,000 years ago         |
| Disappearance of Dinosaurs | 65 mya                   |

[New NCERT Class 12<sup>th</sup> Page No. 123,124,125]

**Q182 Text Solution:**

(D)

ZIFT involves *in-vitro* fertilisation.

[New NCERT Class 12<sup>th</sup> Page No. 48]

**Q183 Text Solution:**

(D)

- In 1953, S.L. Miller, an American scientist created similar conditions in a laboratory scale. He created electric discharge in a closed flask containing CH<sub>4</sub>, H<sub>2</sub>, NH<sub>3</sub> and water vapour at 800 °C.
- He observed formation of amino acids. In similar experiments others observed, formation of sugars, nitrogen bases, pigment and fats.

[New NCERT Class 12<sup>th</sup> Page No. 111]

**Q184 Text Solution:**

(B)

- The release of oxytocin from the maternal pituitary, acts on the uterine muscle and causes stronger uterine contractions, which in turn stimulates further secretion of oxytocin.
- The stimulatory reflex between the uterine contraction and oxytocin secretion continues resulting in stronger and stronger contractions.
- This leads to expulsion of the baby out of the uterus through the birth canal.

[New NCERT Class 12<sup>th</sup> Page No. 38]

**Q185 Text Solution:**

(A)

- Cleavage occurs as the zygote moves through isthmus towards uterus.
- Endometrium is an innermost glandular layer of uterus.

[New NCERT Class 12<sup>th</sup> Page No. 30,36]

**Q186 Text Solution:**

(C)

The first non-cellular forms of life could have originated 3 billion (3000 millions) years back. About 2000 million years ago (mya) the first cellular forms of life appeared on earth.

Difference in appearance of first non-cellular forms of life and appearance of first cellular forms of life = 3000 – 2000 = 1000 million years.

[New NCERT Class 12<sup>th</sup> Page No. 111,112]

**Q187 Text Solution:**

(C)

- Coitus interruptus is natural contraceptive method in which the male partner withdraws his penis from the vagina just before ejaculation so as to avoid insemination.
- Saheli is a new oral contraceptive for the females was developed by scientists at Central Drug Research Institute (CDRI) in Lucknow, India.

[New NCERT Class 12<sup>th</sup> Page No.44, 45]

**Q188 Text Solution:**

(B)

Gonorrhoea, syphilis, genital herpes, chlamydia, genital warts, trichomoniasis, hepatitis-B and HIV leading to AIDS are some of the common sexually transmitted diseases.

[New NCERT Class 12<sup>th</sup> Page No. 47]

**Q189 Text Solution:**

(A)

Five factors are known to affect Hardy-Weinberg equilibrium. These are gene migration or gene flow, genetic drift, mutation, genetic recombination and natural selection.

[New NCERT Class 12<sup>th</sup> Page No. 121]

**Q190 Text Solution:**

(B)





In GIFT, AI and IUI there is *in-vivo* fertilisation takes place, while in ZIFT, ICSI and IUT *in-vitro* fertilisation takes place.

[New NCERT Class 12<sup>th</sup> Page No. 48]

**Q191 Text Solution:**  
(D)

- In adults, each testis is oval in shape.
- Each testis has 250 compartments.
- Each testicular lobule contains one to three highly coiled seminiferous tubules.
- The enlarged end of penis is called glans penis.
- Rete testis is a male sex accessory duct.

[New NCERT Class 12<sup>th</sup> Page No. 27,28]

**Q192 Text Solution:**  
(B)

Copper releasing IUDs increase phagocytosis of sperms within the uterus and the Cu ions released suppress sperm motility and the fertilising capacity of sperms.

[New NCERT Class 12<sup>th</sup> Page No. 44]

**Q193 Text Solution:**  
(D)

- Fossils are remains of hard parts of life-forms found in rocks.
- Jawless fish probably evolved around 350 mya.
- The brain capacities of *Homo habilis* were between 650-800cc.
- The original Darwin finches were seed-eating.

[New NCERT Class 12<sup>th</sup> Page No. 122,125]

**Q194 Text Solution:**  
(A)

- Sertoli cells are found inside seminiferous tubules.
- After spermiogenesis, sperm heads become embedded in the Sertoli cells.

[New NCERT Class 12<sup>th</sup> Page No. 31,32]

**Q195 Text Solution:**  
(D)

- 1 secondary spermatocyte gives rise to 2 spermatozoa and 1 primary oocyte gives rise to 1 ovum.

- 100 secondary spermatocytes will give rise to 200 spermatozoa and 100 primary oocytes will form 100 ova.

[New NCERT Class 12<sup>th</sup> Page No. 32]

**Q196 Text Solution:**  
(C)

|                                               |                                         |
|-----------------------------------------------|-----------------------------------------|
| Heart sound                                   | First sign of growing foetus            |
| Foetus develops limbs and digits              | By the end of second month of pregnancy |
| Formation of major organ system               | By the end of 12 weeks                  |
| First movement and appearance of hair on head | During the fifth month                  |

[New NCERT Class 12<sup>th</sup> Page No. 38]

**Q197 Text Solution:**  
(A)

*Tyrannosaurus rex* was about 20 feet in height and had huge fearsome dagger like teeth.

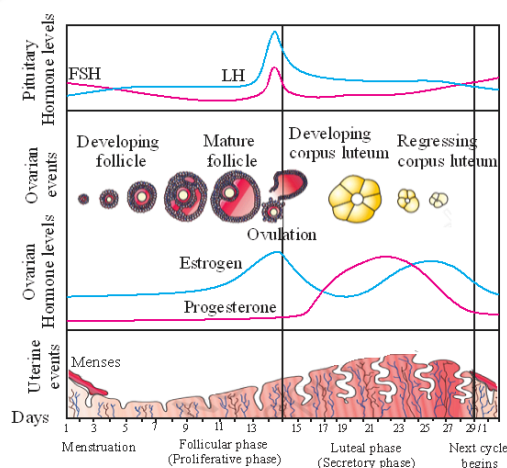
[New NCERT Class 12<sup>th</sup> Page No. 124]

**Q198 Text Solution:**  
(A)

Vertebrate hearts and brains exhibit divergent evolution.

[New NCERT Class 12<sup>th</sup> Page No. 115]

**Q199 Text Solution:**  
(C)



Diagrammatic presentation of various events during a menstrual cycle.

[New NCERT Class 12<sup>th</sup> Page No. 34]

**Video Solution:**





Q200 Text Solution:

(B)

*Ramapithecus* is an earlier hominid, followed by *Australopithecines*, then *Homo habilis*, and finally *Homo erectus*, which is more recent in the evolutionary timeline.

[New NCERT Class 12<sup>th</sup> Page No. 124]



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